Dr. Paul Töchterle

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Profile

Hydrogeologist specialising in alpine water and energy systems.

My work combines fieldwork, statistical analysis and numerical modelling with practical applications in spatial planning. Alongside my technical work, I develop Python-based calculation and analysis tools that translate geological and hydrological methods into reproducible, practice-oriented workflows.

Hydrogeology	**Field Surveys**	**Mapping**
Drinking Water Supply	**Permafrost**	**Modelling & Simulation**
Climate Change	**Geothermal Energy**	**Data Management**

Selected Projects

Hydrogeology of Major Springs in Tyrol Hydrogeology · Data Management · GIS
Field surveys, hydrogeological analysis and water resource assessment of 93 major springs in Tyrol for strategic drinking water supply planning, commissioned by the Tyrolean Provincial Government.

Drinking Water Demand Model for Tyrol Machine Learning · Drinking Water Supply · Data Management
Development of a machine-learning-based demand model as part of the drinking water supply strategy, commissioned by the Water Management Department of the Tyrolean Provincial Government. The model predicts current and future per-capita demand using statistical and climatic indicators and model projections.

Universal Runoff Model for Tyrol Machine Learning · Hydrology · Data Management
Development of a universal rainfall–runoff model for Tyrol using machine learning.

Permafrost Heat Flux Modelling Data Management · Climate Adaptation · Programming
Simplified heat flux model for calculating temperature profiles in permafrost soils. Open web tool

Mount Resilience Data Management · Climate Adaptation · Programming
EU project on climate change adaptation in mountain regions. Analysis of climate data at municipal level and development of an online tool for municipalities.

Glacier Reconstruction in the Stubai Alps GIS · Programming · Glaciers
Web tool for visualising glacier extents during the Little Ice Age in the Stubai Alps. Open web tool

Low-Ex Heating Network, Auffach / Wildschönau Spatial Energy Planning · Hydrogeology · Stakeholder Management
Feasibility study for a low-temperature heating network using spring water as a thermal energy source.

Municipal Energy Supply Strategies Spatial Energy Planning · GIS · Data Management
Analysis of municipal energy demand and available resources to support planning for local heating and energy supply.

Seasonal Performance Factor Calculator (Heat Pumps) Programming · Climate Data
Web application for predicting the seasonal performance factor of air-source heat pumps using the bin method and local climate data. Open web tool

Professional Experience

2024 – present
Energieagentur Tirol

Project Manager

Concept development and feasibility studies in drinking water supply, small hydropower, municipal spatial energy planning and geothermal energy. Hydrological and hydrogeological fieldwork, spatial analysis and rainfall–runoff modelling as a methodological basis for planning decisions.

Development of internal Python-based analysis and calculation tools and web applications for use within Energieagentur Tirol. Establishment of standardised, reproducible analysis workflows and an internal Python codebase for use across projects.

Spatial Energy Planning · **Hydrogeology** · **Python** · **QGIS / ArcGIS**

2019 – 2024
University of Innsbruck

University Research and Teaching Assistant

Research on permafrost and climate change: laboratory work, expeditions and fieldwork, numerical modelling, and independent teaching of GIS and spatial statistics at BSc and MSc levels.

Research · **Numerical Modelling** · **Independent Teaching** · **Time Series Analysis**

2018 – 2020
KIT – Campus Garmisch-Partenkirchen

Laboratory Manager

Management of the Stable Isotope Laboratory. CHNO isotope analysis in biogeochemical systems, method development and automation of measurement procedures.

Laboratory Techniques · **Isotope Analysis** · **Automation**

2014 – 2018
University of Innsbruck

Laboratory Assistant

Analytical work and maintenance in the laboratories of the Quaternary Geology Research Group; fieldwork on various projects (cave expeditions, percussion core sampling, and hydrological and hydrogeological measurements).

2008 – 2012
Self-employed

Freelance Sound Engineer

Concerts, conference audio, theatre and recording studio work.

Education

2019 – 2023
Univ. of Innsbruck

PhD in Earth Sciences

Doctoral thesis: Permafrost response to climate change during the last glacial period

HYPO Tirol Doctoral Thesis Award 2024

2015 – 2018
Univ. of Innsbruck

MSc in Earth Sciences

Master's thesis: Cryogenic cave carbonates from the Ural Mountains (Russia)

2012 – 2015
Univ. of Innsbruck

BSc in Earth Sciences

2004 – 2006
SAE Munich

Diploma in Audio Engineering

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Skills & Tools

PROGRAMMING	
Python – Scientific Stack (NumPy, Pandas, SciPy, Xarray)	Expert
Python – Geospatial Stack (GDAL, GeoPandas, Cartopy)	Expert
Python – Web Apps (Streamlit, Flask)	Advanced
Python – ML / Modelling (Scikit-learn, PyTorch)	Advanced
SQL	Advanced
HTML / CSS	Basic
R, Matlab	Basic
GIS & CAD	
QGIS	Expert
ArcGIS Pro	Expert
AutoCAD	Basic
OFFICE & GRAPHICS	
MS Office	Expert
Corel Draw	Expert
Agisoft Metashape	Basic
GIMP / Inkscape	Basic
LANGUAGES	
German	Native
English	Fluent
Italian	Basic
Dutch	Basic

Expeditions & Fieldwork

Tyrol 2024 – present	Hydrogeological fieldwork for strategic drinking water supply projects commissioned by the Tyrolean Provincial Government.
Great Britain 2019 – 2022	Sampling and surveying in the karst regions of England (Peak District, Mendip Hills) as part of doctoral research.
Northeast Greenland 2019	Multi-week expedition to Kronprins Christian Land (Greenland Caves Project, Prof. G.E. Moseley): cave exploration, stratigraphy of Palaeozoic carbonate sequences, and mapping of Late Glacial moraines.
Bavaria 2018 – 2019	Soil science projects with the Biogeochemical Cycles Research Group at KIT Garmisch-Partenkirchen.
Russia (Urals) 2017	Ural-Expedition der AG Quartärgeologie (Dr. Y. Dublyansky): Erkundung und Beprobung von Höhlen in den Karstgebieten der Regionen Perm, Denezhkin Kamen und Bashkortostan.
Japan 2016	Forschungsfahrt SO-251 (FS Sonne): Japan-Graben und Nankai-Trog. Porenwasserchemie und Chemo-Stratigraphie von Sedimentkernen.
Österreich 2014 – 2019	Feldforschungsprojekte der AG Quartär (Prof. Christoph Spötl): Höhlenerkundungen, Rammkernsondierungen, hydrogeologische Feldarbeiten.

Peer-Review Publikationen (Auswahl)

- **Töchterle, P.**, Baldo, A., Murton, J. B., et al.: Reconstructing Younger Dryas Ground Temperature and Snow Thickness from Cave Deposits. *Climate of the Past* 20, 2024. doi:10.5194/cp-20-1521-2024
- **Töchterle, P.**, Steidle, S., Edwards, R. L., et al.: 230Th/U Isochron Dating of Cryogenic Cave Carbonates. *Geochronology* 4, 2022. doi:10.5194/gchron-4-617-2022
- **Töchterle, P.**, Yang, F., Rehschuh, S., et al.: Hydraulic Water Redistribution by Silver Fir occurring under Severe Soil Drought. *Forests* 11, 2020. doi:10.3390/f11020162
- **Töchterle, P.**, Dublyansky, Y., Stöbener, N., et al.: High-resolution isotopic monitoring of cave air CO_2. *Rapid Communications in Mass Spectrometry* 31, 2017. doi:10.1002/rcm.7859
- Seelig, S., Thalheim, F., Seelig, M., **Töchterle, P.**, et al.: Young water fractions in spring discharge. *Journal of Hydrology* 670, 2026. doi:10.1016/j.jhydrol.2026.135221
- Donner, A., **Töchterle, P.**, Spötl, C., et al.: Cryogenic cave minerals recorded the 1889 CE melt event in northeastern Greenland. *Climate of the Past* 19, 2023. doi:10.5194/cp-19-1607-2023
- Johnston, V. E., Borsato, A., Frisia, S., **Töchterle, P.** et al.: Evidence of thermophilisation during the Last Interglacial in the Italian Alps. *Scientific Reports* 8, 2018. doi:10.1038/s41598-018-21027-3
- Vázquez, E., Teutscherova, N., **Töchterle, P.** et al.: Gross nitrogen transformations in tropical pasture soils. *Soil Biology and Biochemistry* 151, 2020. doi:10.1016/j.soilbio.2020.108058

Vollständige Publikationsliste: Google Scholar · Researchgate

Lehrtätigkeit

MSc-Niveau

- Einführung in Geographische Informationssysteme – ArcGIS Pro & QGIS (Univ. Innsbruck, 2020, 2021)
- Räumliche Daten in den Erdwissenschaften – Zeitreihenanalyse, Interpolation, Clusteranalyse (Univ. Innsbruck, 2022, 2023)
- Quartärgeologisches Geländepraktikum – Kartierung glazio-lakustriner Ablagerungen (Univ. Innsbruck, 2017, 2020–2023)

BSc-Niveau

- Einführung in die Geländearbeit – Karbonat-Sedimentologie der Nördlichen Kalkalpen (Univ. Innsbruck, 2019–2023)